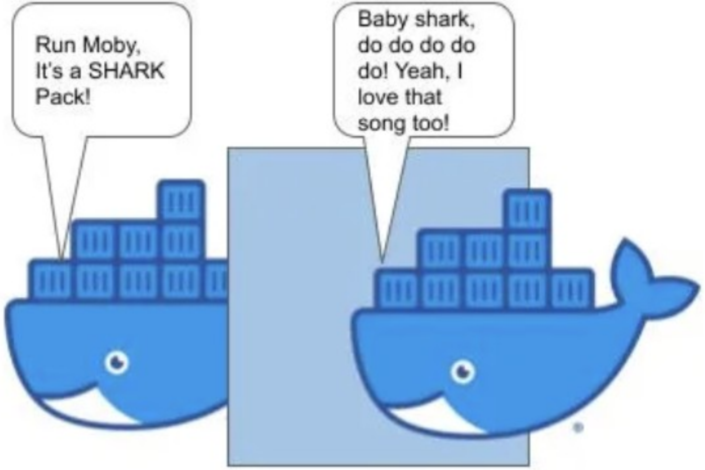
Kubernetes + ML Deployment

Recap

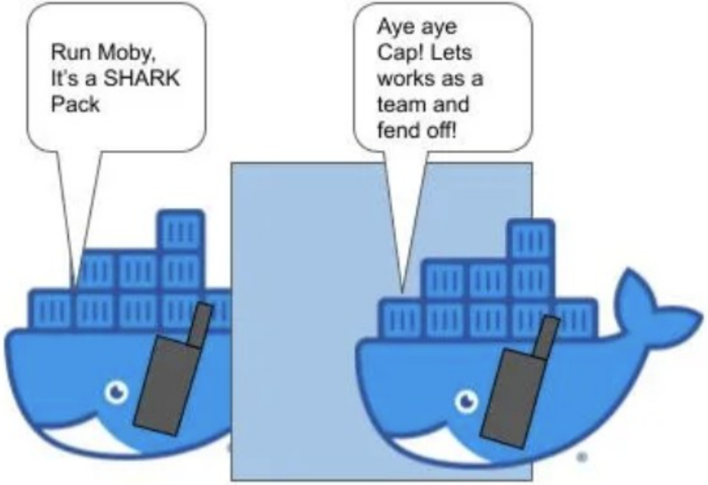
- Cloud
- Virtualization
- Containers
- Docker
- Docker Images
- Docker Networking
- Kubernetes

```
docker tag ml-app:v1 quay.io/ppnaik/ml-app:v1

docker push quay.io/myorg/ml-app:v1
```



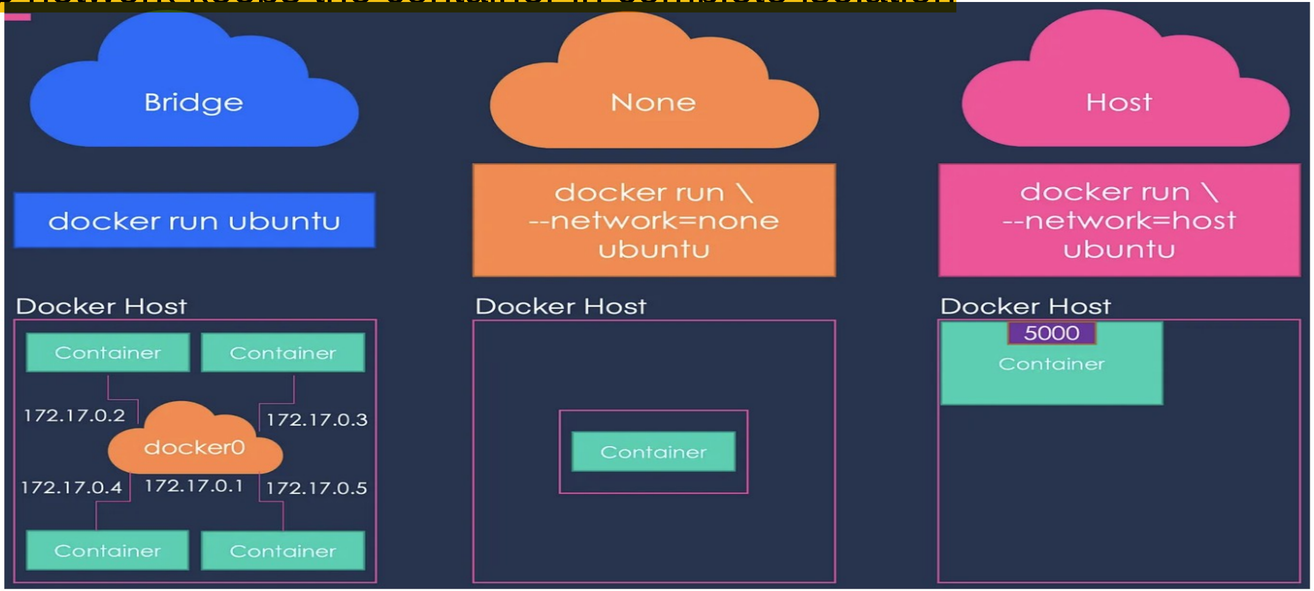
Barrier to communication between docker containers



Communications using the Network Bridge

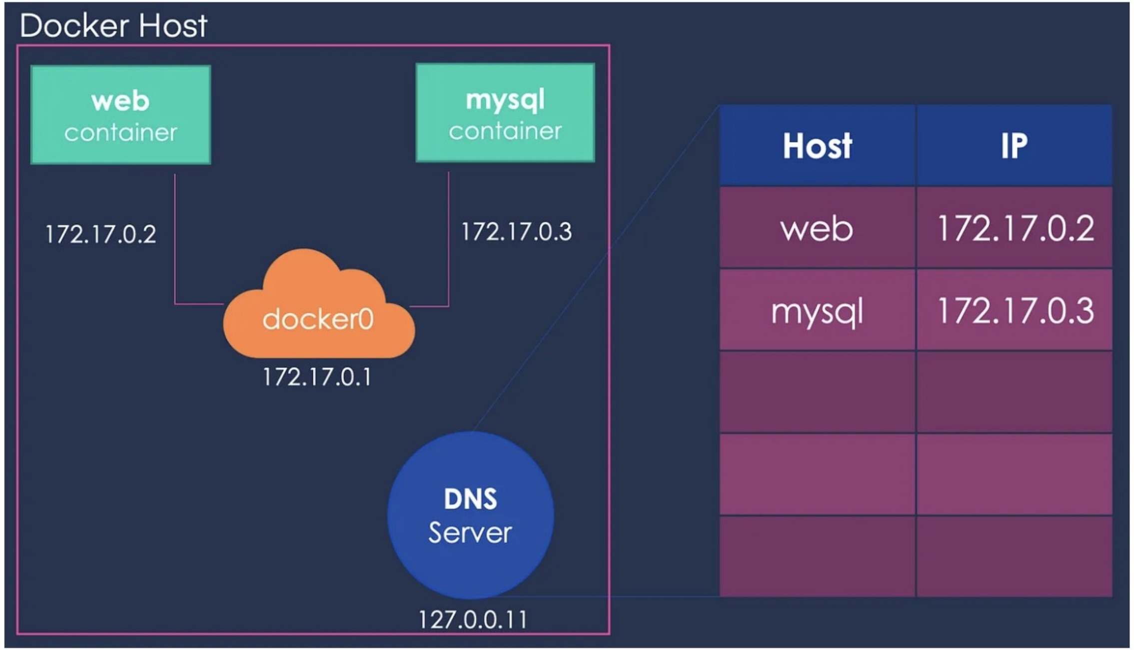
Docker Networking

- Three networks created automatically - Bridge, Host, and None
- The **Bridge** network assigns IPs in the range of *172.17.x.x* to the containers
- Host Mode: Accessible on the same port on the docker host without any explicit port mapping
- **None** network keeps the container in complete isolation

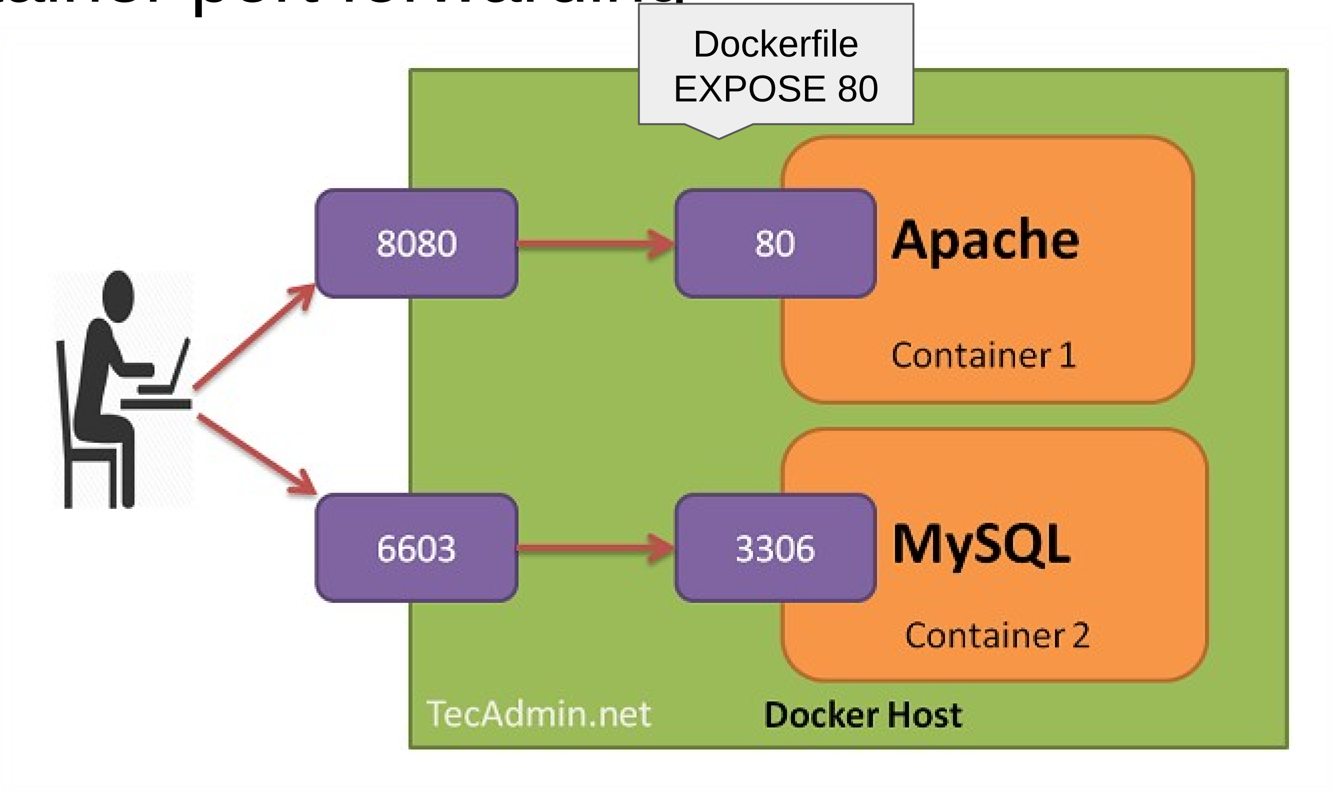


Docker Bridge Network

- Default network container connect to bridge



Container port forwarding



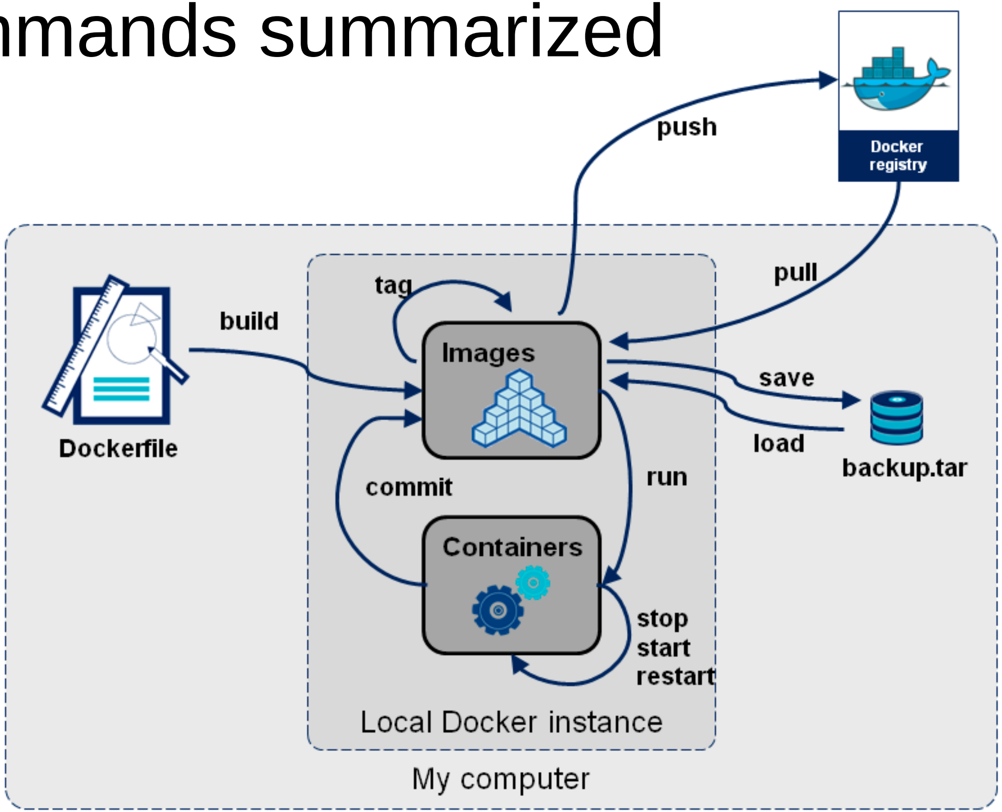
https://tecadmin.net/tutorial/wp-content/uploads/2017/09/Docker-port-binding.png

Port forwarding

```
docker run -d --name mycontainer1 -p 8080:80 mypython-app:v1
```

See in action: go to browser and enter URL localhost:8080

Docker commands summarized

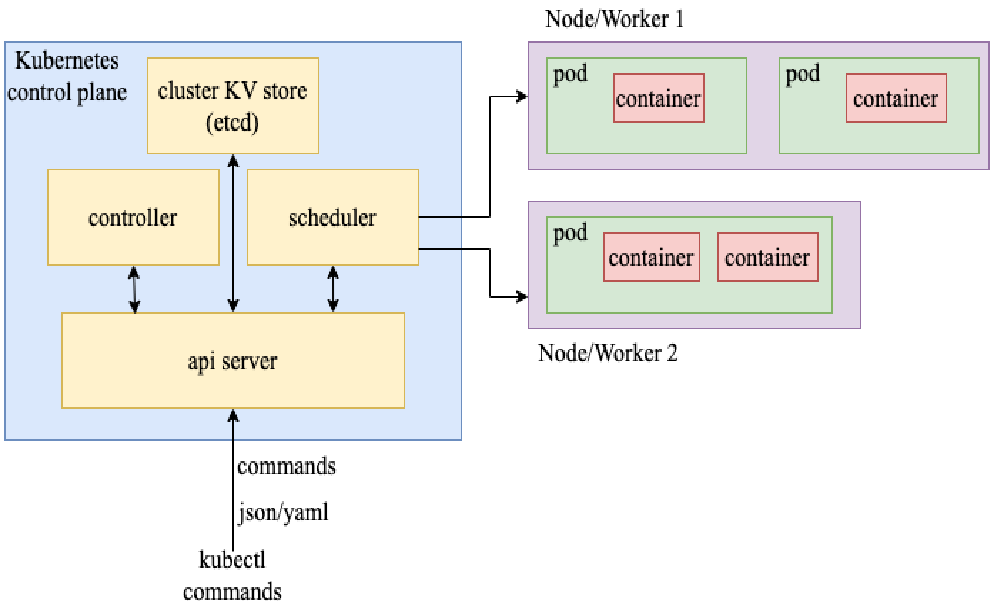


https://blog.octo.com/docker-registry-first-steps/

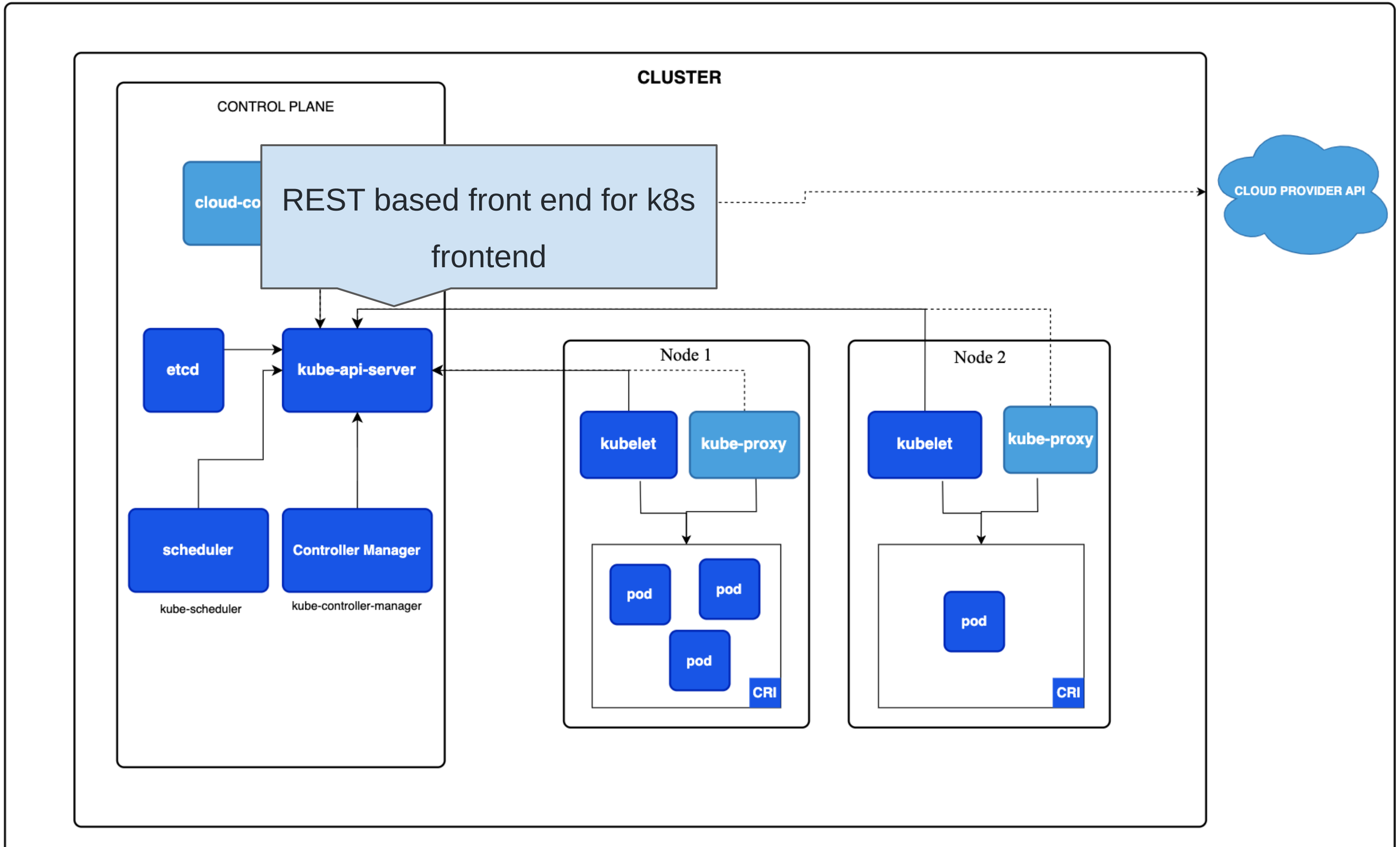
Container orchestration

What is kubernetes?

- Docker: Single machine container deployment
- Kubernetes (k8s): Container Orchestration
  - Across a cluster of machines
  - Manage automated deployment, scaling

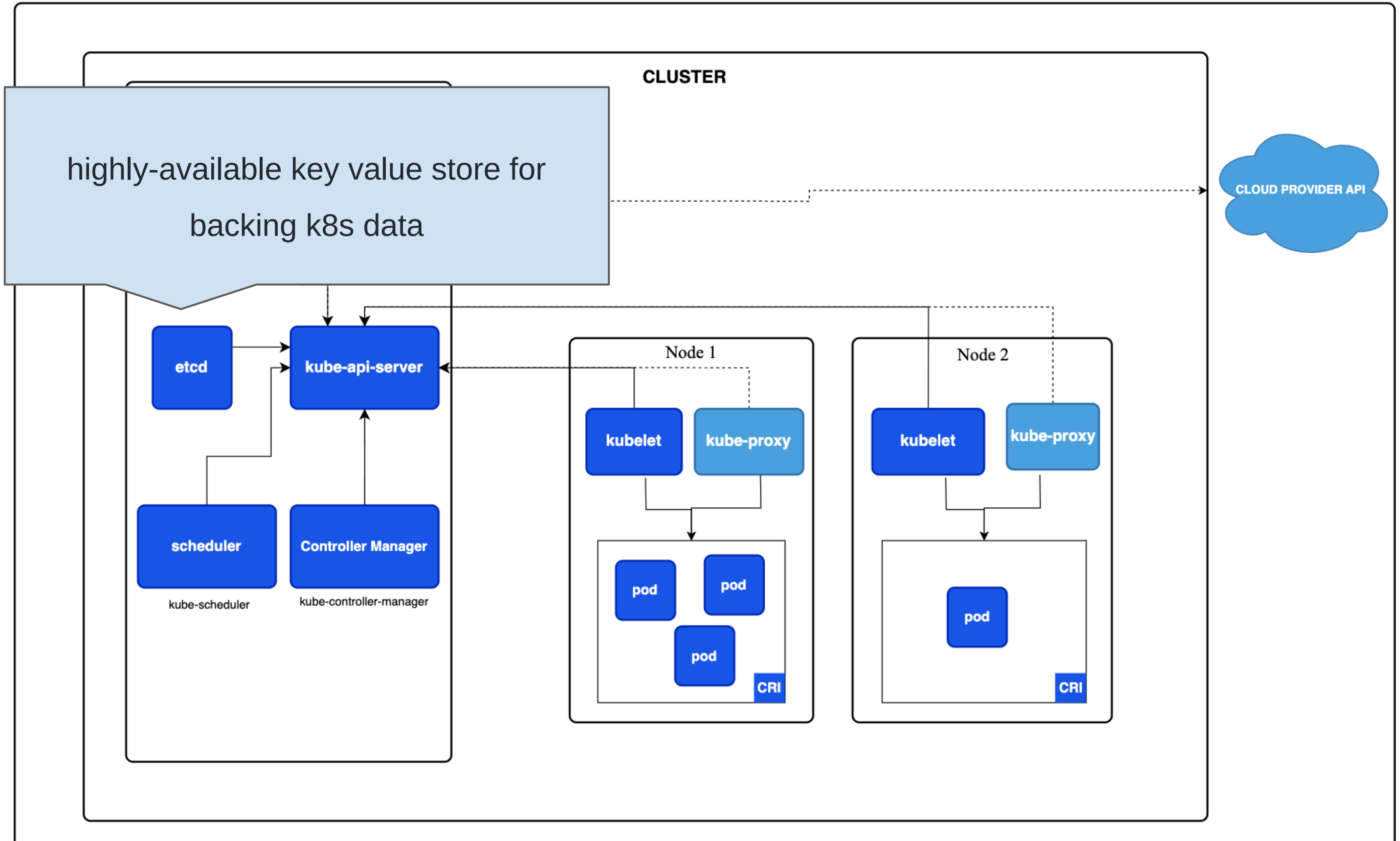


Kubernetes Architecture

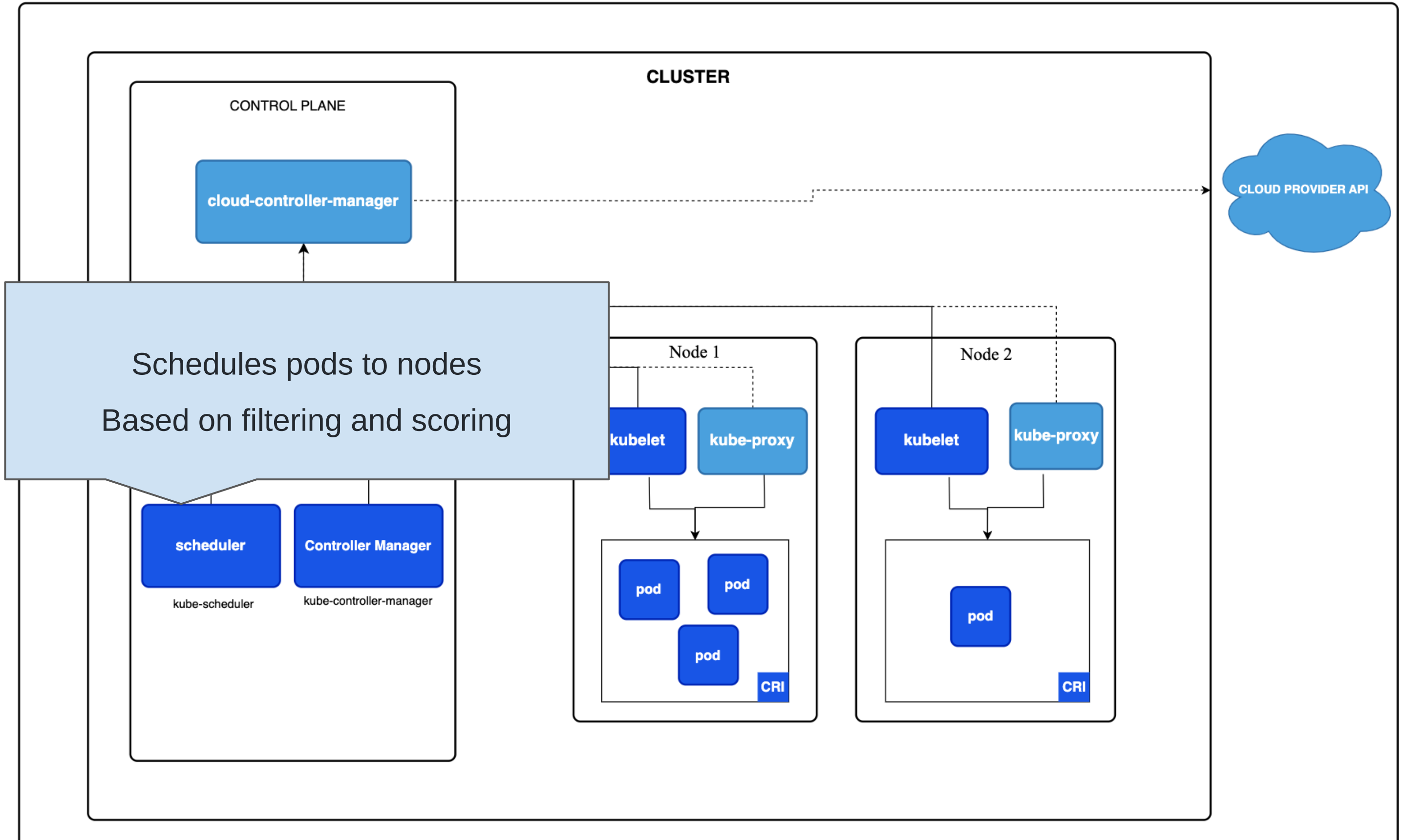


Kubernetes Architecture

highly-available key value store for backing k8s data



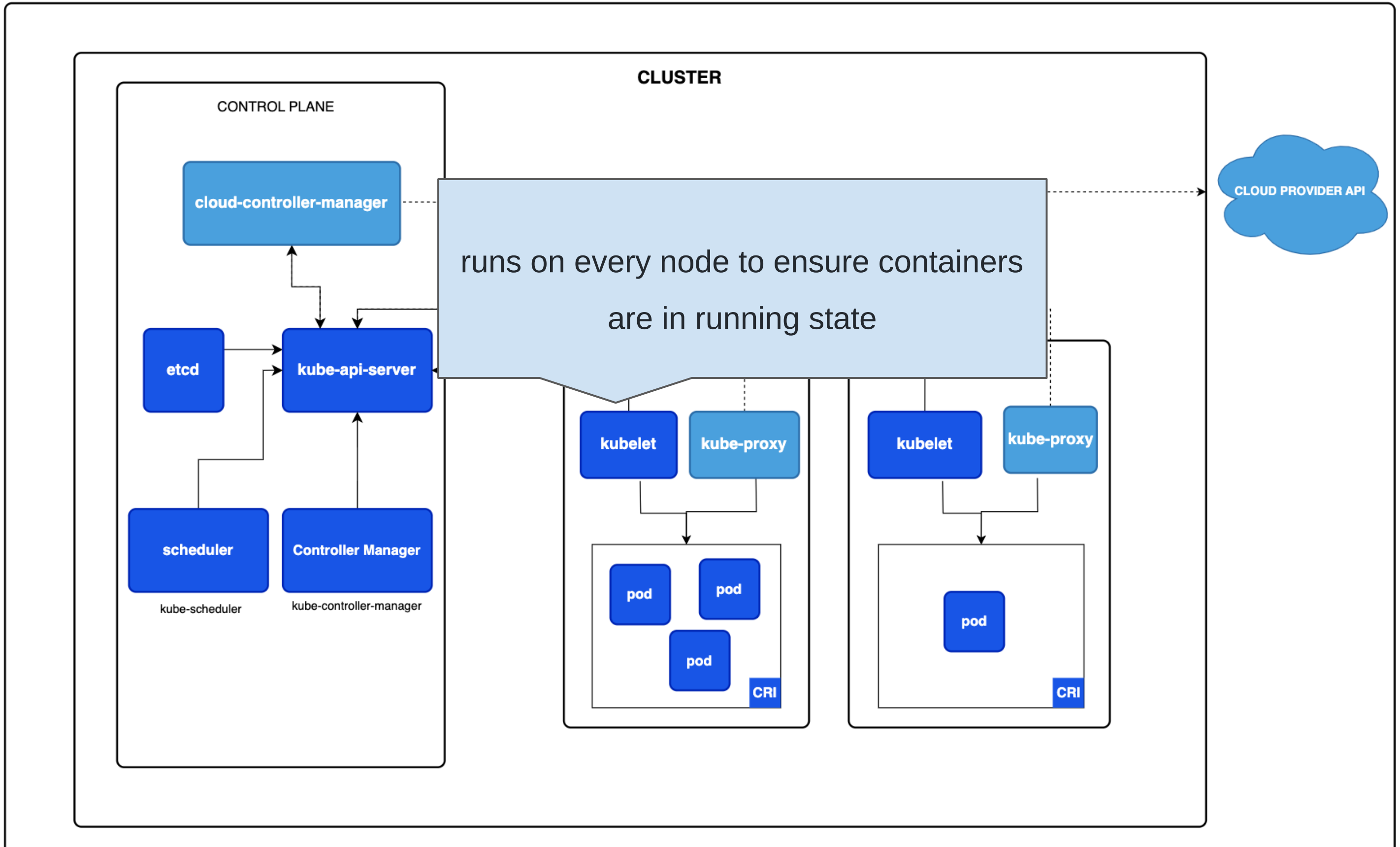
Kubernetes Architecture



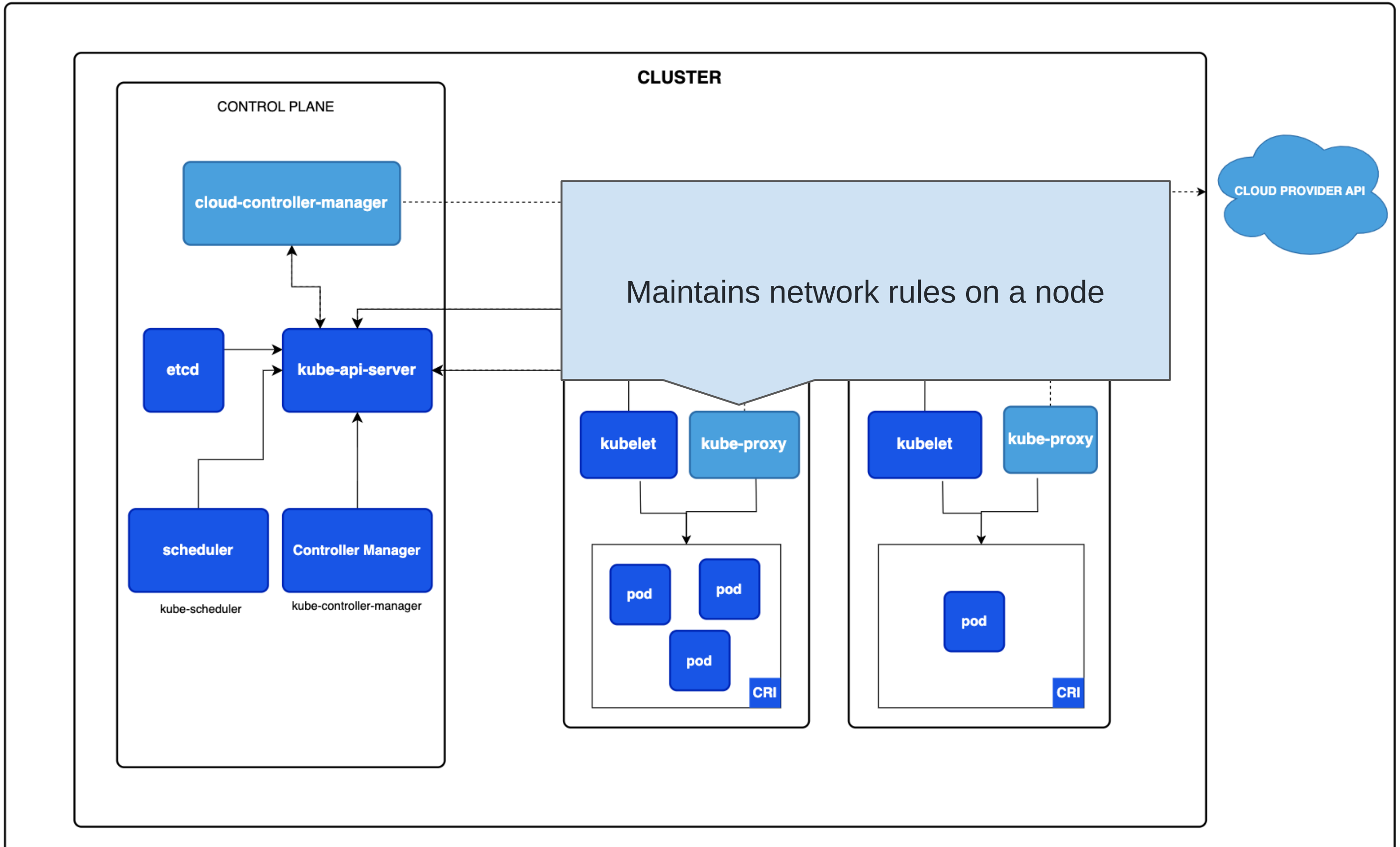
Schedules pods to nodes

Based on filtering and scoring

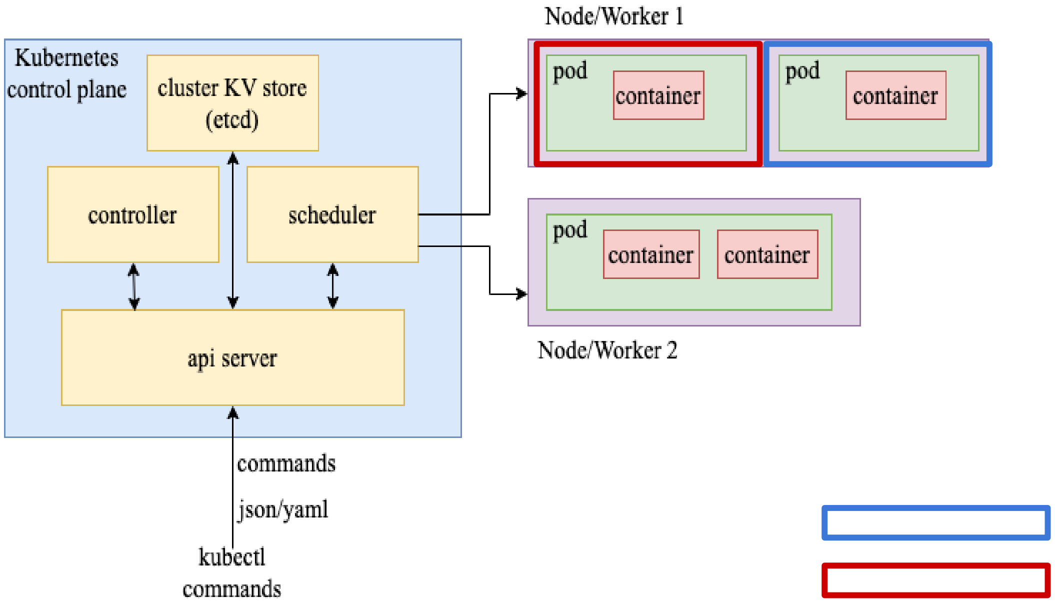
Kubernetes Architecture



Kubernetes Architecture



Namespace



Namespace: Isolated environment for each user

Kubectl get started

- Create a namespace
- `kubectl create namespace <urname>`
- Go to the namespace:
- `kubectl config set-context --current --namespace=<namespacename>`

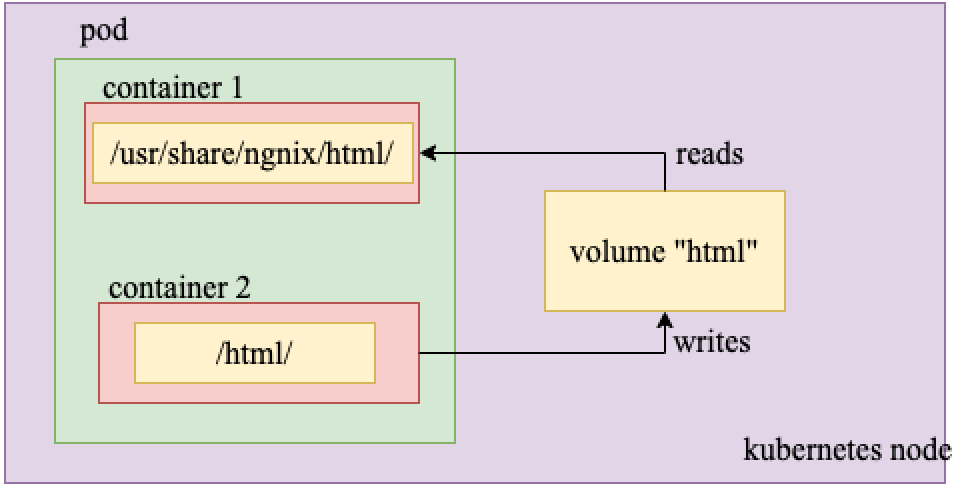
To test locally:

- Install minikube
- Install kubectl
- Install docker/podman
- `minikube start --driver=docker`
- `minikube kubectl create namespace abc`
- `minikube kubectl -- config set-context --current --namespace=abc`
- `alias kubectl="minikube kubectl --"`

Pods

Application specific logical host.

- single unit of deployment
- group of containers with shared storage and network resources.



Pod Example add namespace

- Create a single nginx pod:
- kubectl apply -f ngnix-pod.yaml -n <namespace>
- Check the pod:
  - kubectl describe pod nginx
  - Kubectl exec -it nginx -- /bin/ba

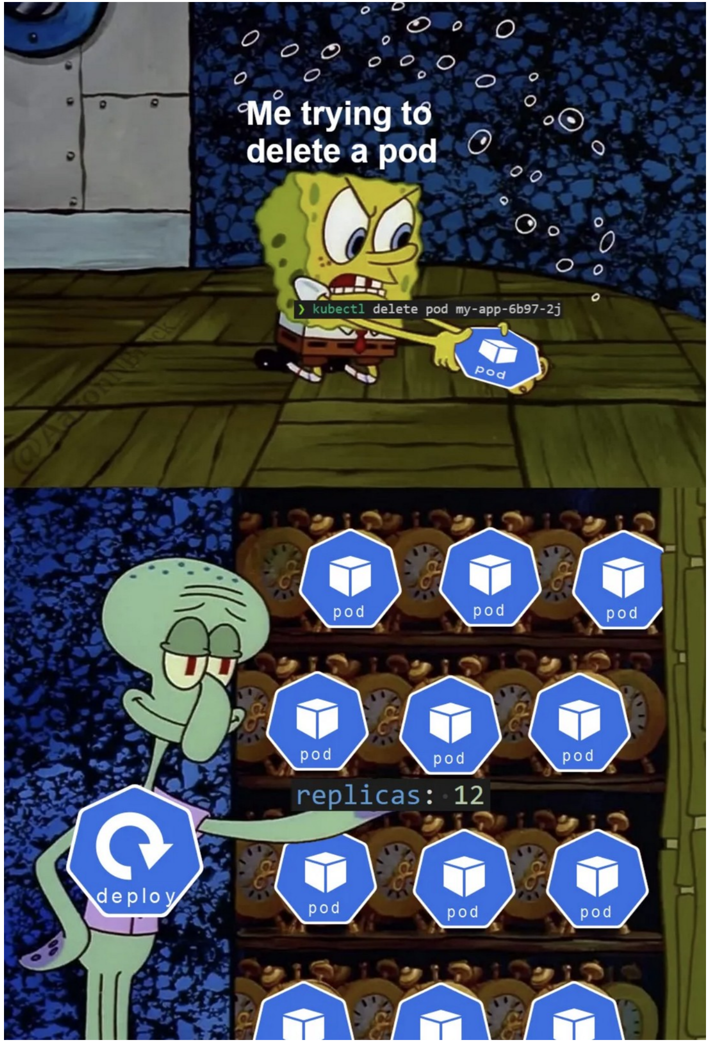
```yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx
spec:
  containers:
    - name: nginx
      image: nginx
      ports:
        - containerPort: 80
```

Another Pod Yaml Example

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: netshoot
spec:
  containers:
    - name: netshoot
      image: nicolaka/netshoot
      command: ["/bin/bash"]
      args: ["-c", "while true; do ping 172.30.227.43; sleep 60;done"]
```

Deployments

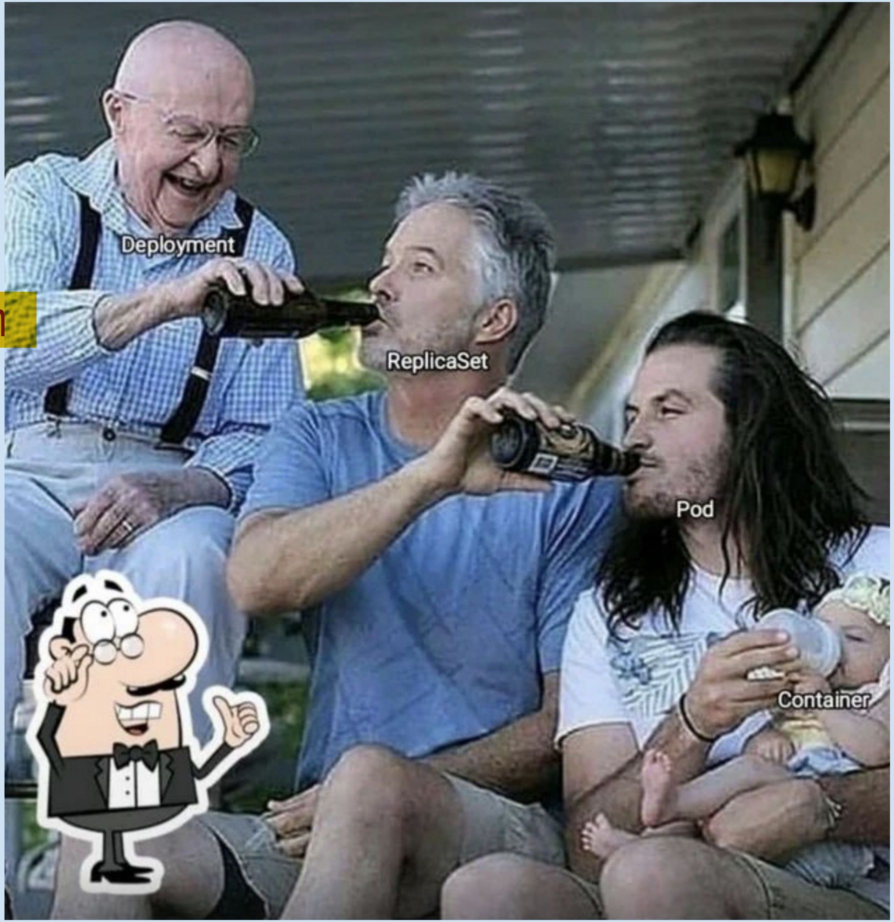
- Kubernetes Controllers
- Features:
  - Rolling Update
  - Rollback
  - Version Control



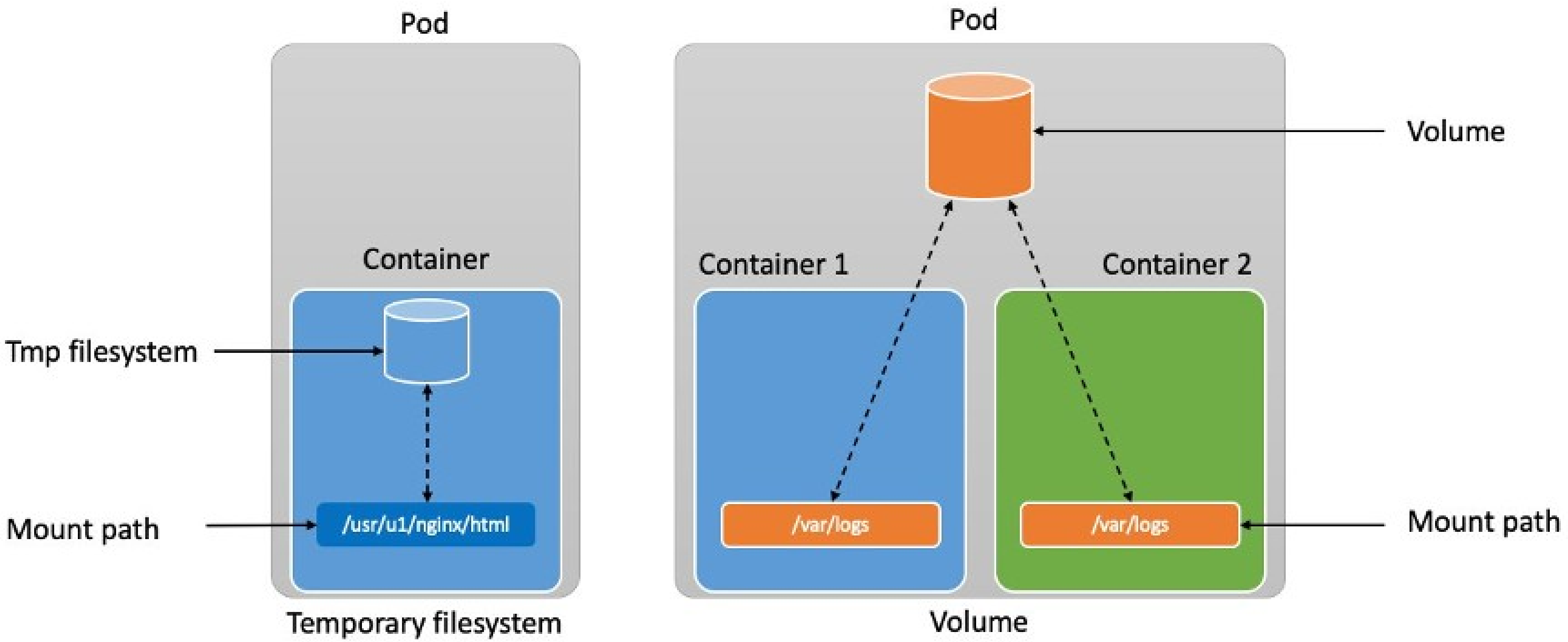
```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

Deployment Example

- `kubectl apply -f nginx_dep.yam`
- Replicas, delete and see them start



Persistent Volume



Persistent Volume and PersistentVolumeClaim

- PersistentVolume (PV):
  - is a cluster resource
  - Implemented as plugin (hostpath,local,nfs)
  - Have a lifecycle independent of a pod
- PersistentVolumeClaim (PVC) :
  - is a storage request from a user.
  - Request for volume similar to CPU, memory
  - PVC access modes: ReadWriteOnce, ReadOnlyMany, ReadWriteMany, or ReadWriteOncePod

Statefulset and Daemonset

Statefulset

- Deployment and scaling of stateful pods
- Stateful pod: requires persistent storage

Daemonset

- An instance of the pod runs on each node
- System daemons and background processes

https://semaphoreci.com/blog/replicaset-statefulset-daemonset-deployments

Services

- Abstraction for an application running multiple replica
- Service discovery to find matching endpoint
- Multi-port Service
- Type:
  - ClusterIP:
  - NodePort:
  - LoadBalancer:
  - ExternalName:
- Service Discovery

Types of Service

- ClusterIP:
  - Default service type. Accessible within the cluster
- NodePort:
  - Allocates a port on each node to forward request to the service
  - `--service-node-port-range` flag (default: 30000-32767)
- Load Balancer
  - Traffic from the external load balancer is directed at the backend Pods.
  - You can specify a `loadBalancerIP` or use ephemeral IP
- ExternalName
  - Service to a DNS name
  - `my.database.example.com`

Service

- `kubectl apply -f nginx-service.yaml`
- `kubectl describe service nginx-service`
- Check connectivity (use the netshoot pod)
  - `curl <serviceIP:port>`
  - Which IP, which port ??

```yaml
apiVersion: v1
kind: Service
metadata:
 name: nginx-service
spec:
 selector:
   app: nginx
 ports:
   - port: 8765
     targetPort: 80
 type: LoadBalancer
```

Ml-app example

```
kubectl apply -f configmap.yaml
kubectl apply -f pv.yaml

kubectl apply -f deployment.yaml

kubectl port-forward -n ml-app svc/ml-app 8002:80
```

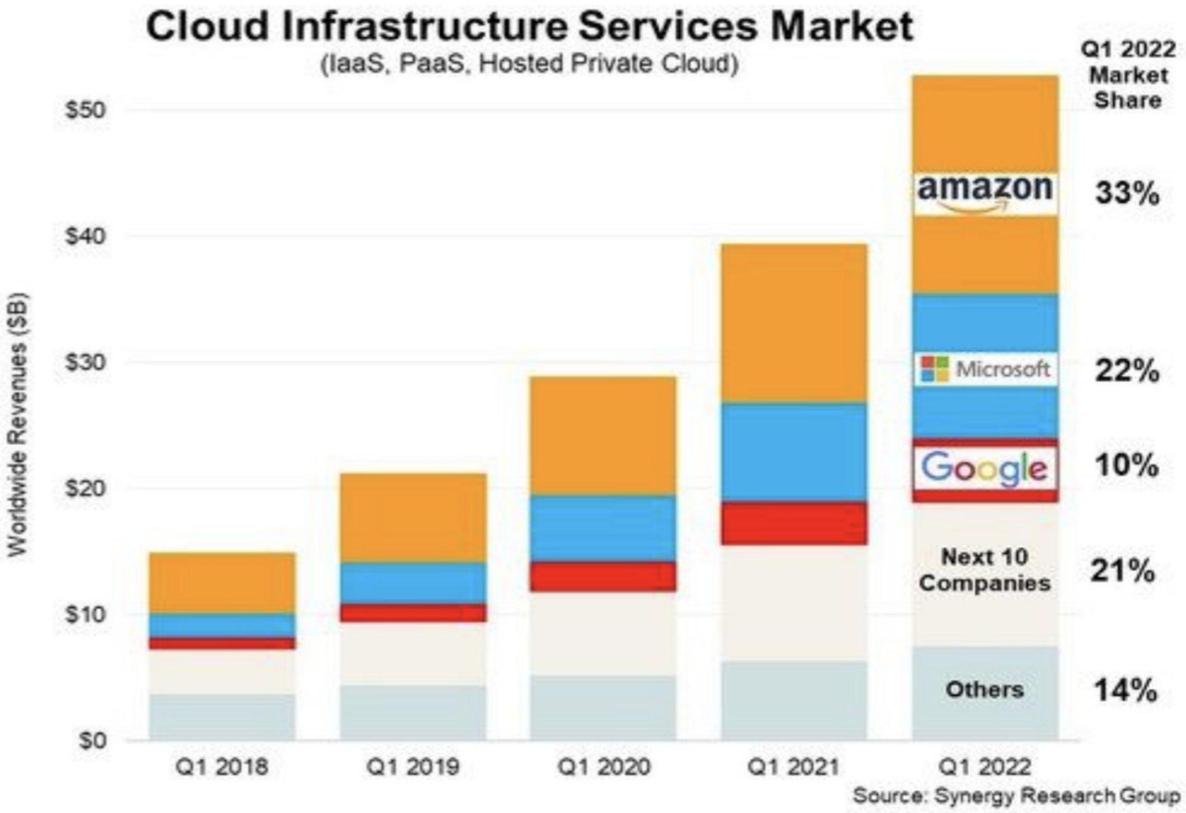
```
curl -X POST http://localhost:8002/predict \
  -H "Content-Type: application/json" \
  -d '{"texts": ["This movie was amazing!"]}'
```

Try it out

https://github.com/pnl-iiitd/acm_fsn/tree/main (k8s and service)

https://medium.com/aspnetrun/deploying-microservices-on-kubernetes-35296d369fdb

Cloud Market



Hyperscale Cloud Providers

	AWS EC2	Azure	GCP
Providers	Amazon	Microsoft	Google
Strengths	Market leader/ Pricing Model	Integration with Microsoft suite	Fastest growing for AI-ML market
Users	Netflix, Airbnb, Coursera	HSBC, Starbucks, HP	Paypal, Unilever, Toyota
Scale	200 Data centers (22 regions)	More regions (66)	26 regions, Low latency

Hybrid Cloud

- Public Cloud
- Private Cloud/ On Prem

Multi-Cloud